



US 20250279513A1

(19) **United States**

(12) **Patent Application Publication**
LEE et al.

(10) **Pub. No.: US 2025/0279513 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **BATTERY CELL INCLUDING SUPPORT PORTION**

(52) **U.S. Cl.**

CPC **H01M 50/169** (2021.01); **H01M 50/103** (2021.01); **H01M 50/15** (2021.01)

(71) Applicant: **SK On Co., Ltd.**, Seoul (KR)

(72) Inventors: **Dong Hyeon LEE**, Daejeon (KR);
Hyung Ju PAIK, Daejeon (KR); **Do Won KIM**, Daejeon (KR)

(57)

ABSTRACT

(21) Appl. No.: **18/983,357**

(22) Filed: **Dec. 17, 2024**

(30) **Foreign Application Priority Data**

Mar. 4, 2024 (KR) 10-2024-0030521

Publication Classification

(51) **Int. Cl.**

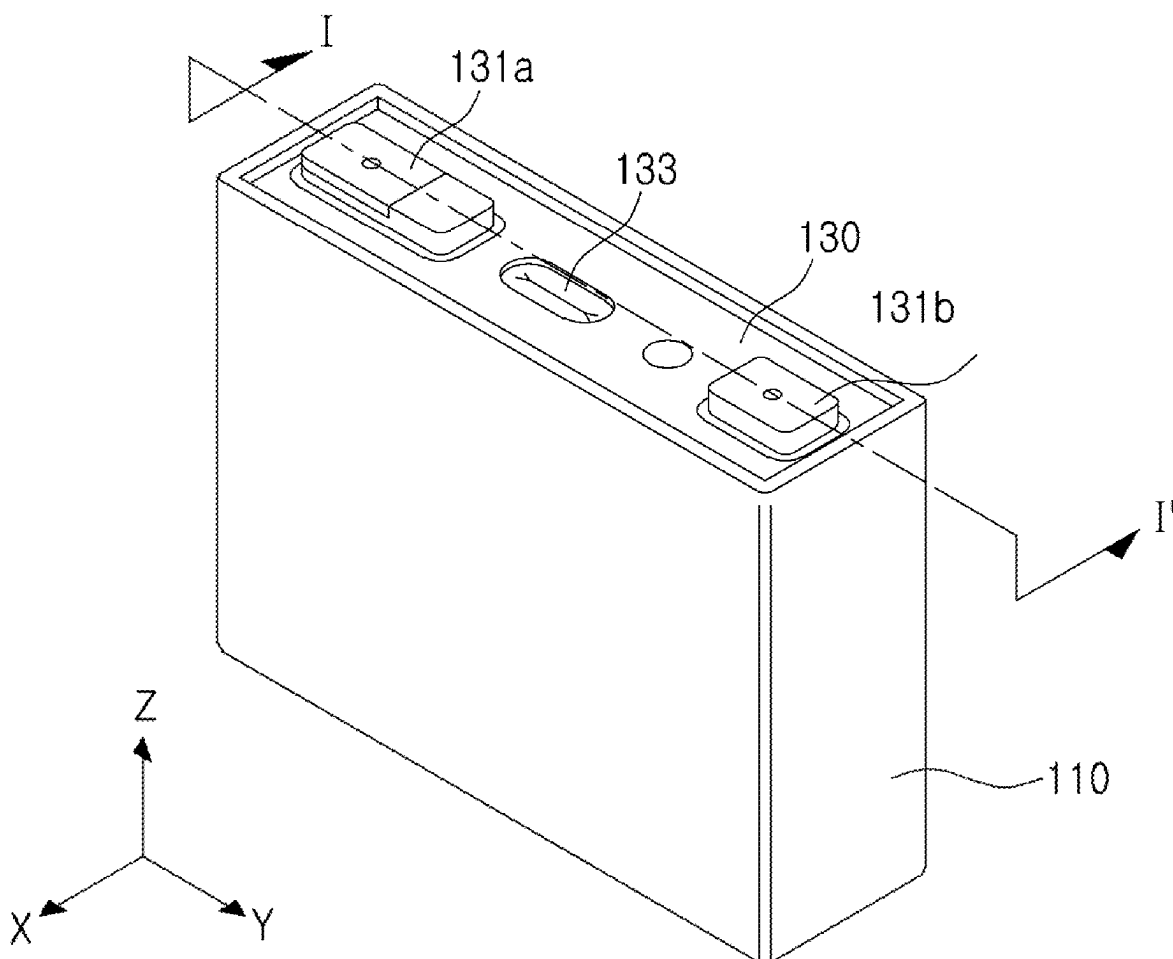
H01M 50/169 (2021.01)

H01M 50/103 (2021.01)

H01M 50/15 (2021.01)

A battery cell includes a case accommodating an electrode assembly in an internal accommodation space and having an opening formed therein, a cap plate at least partially inserted into the accommodation space through the opening and coupled to the case, and a support portion disposed in the accommodation space and supporting a lower surface of the cap plate, wherein a side surface of the cap plate facing the internal surface of the case is spaced apart 10 from the internal surface of the case by a predetermined distance to form a fixing space, and the support portion is disposed below the cap plate so that at least a portion thereof forms a lower surface of the fixing space.

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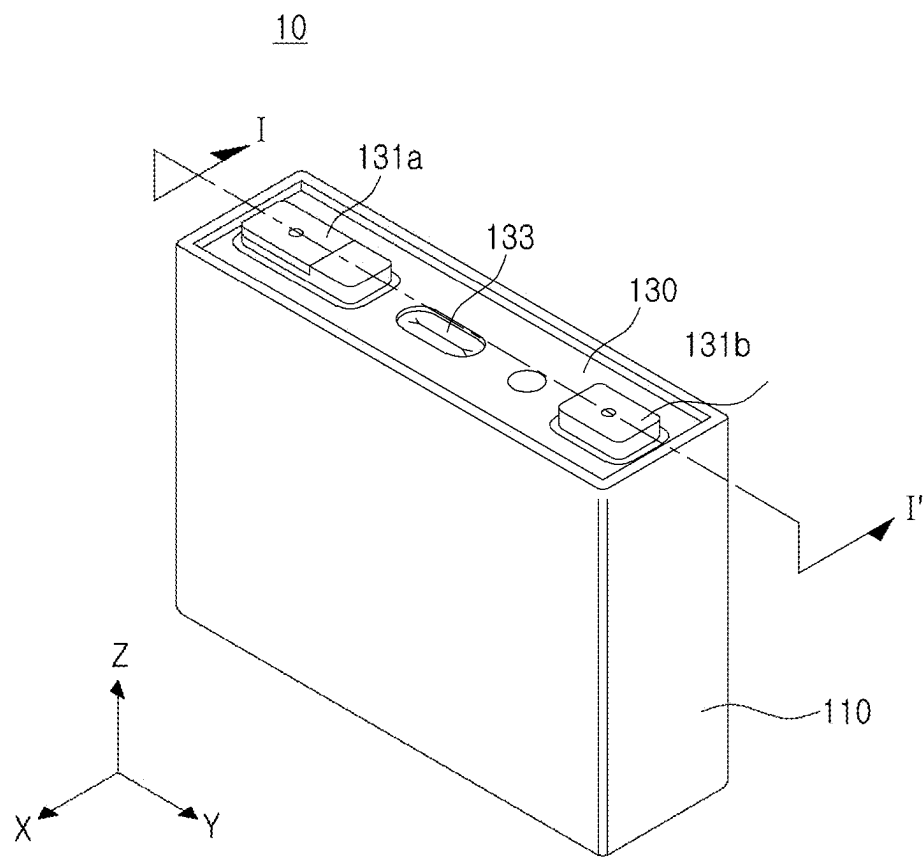


FIG. 1

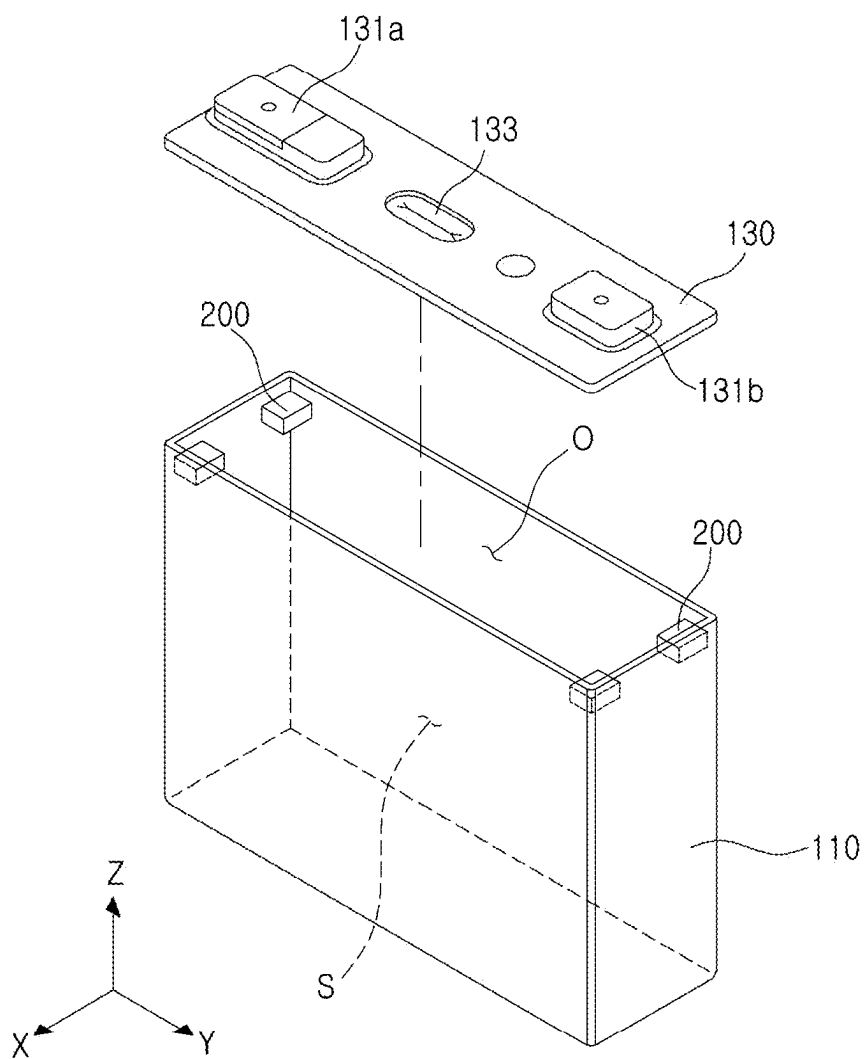


FIG. 2A

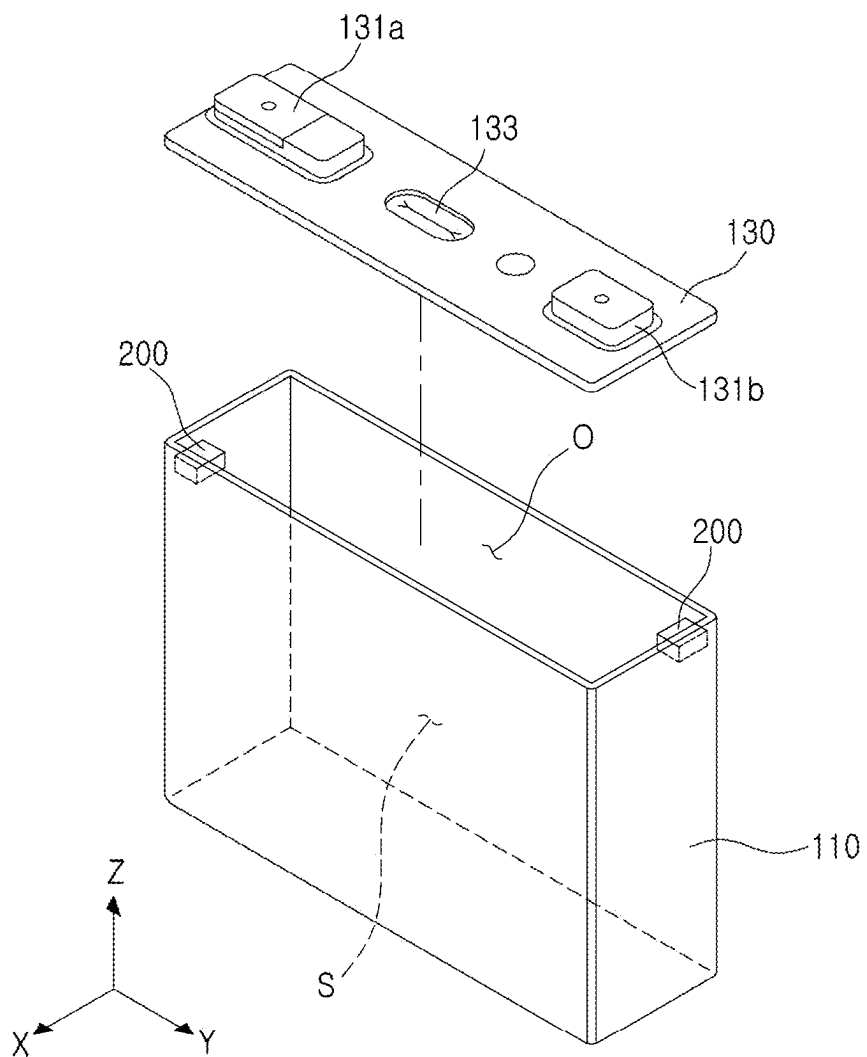


FIG. 2B

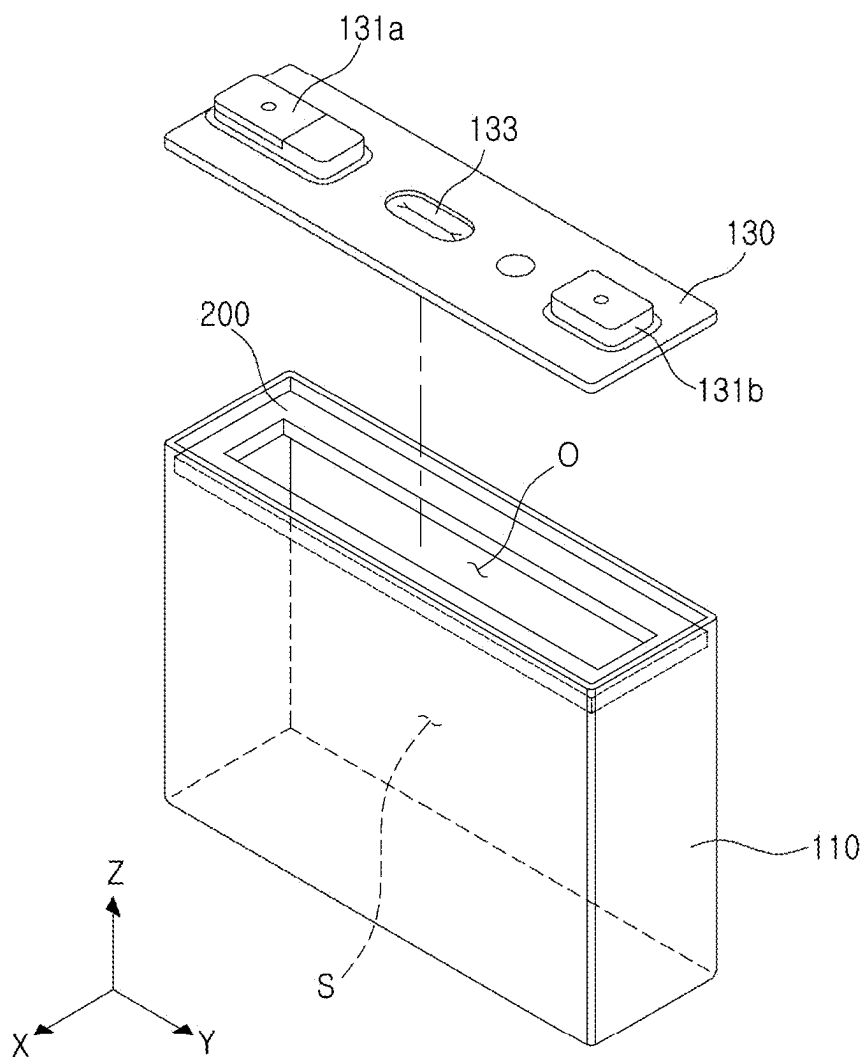


FIG. 2C

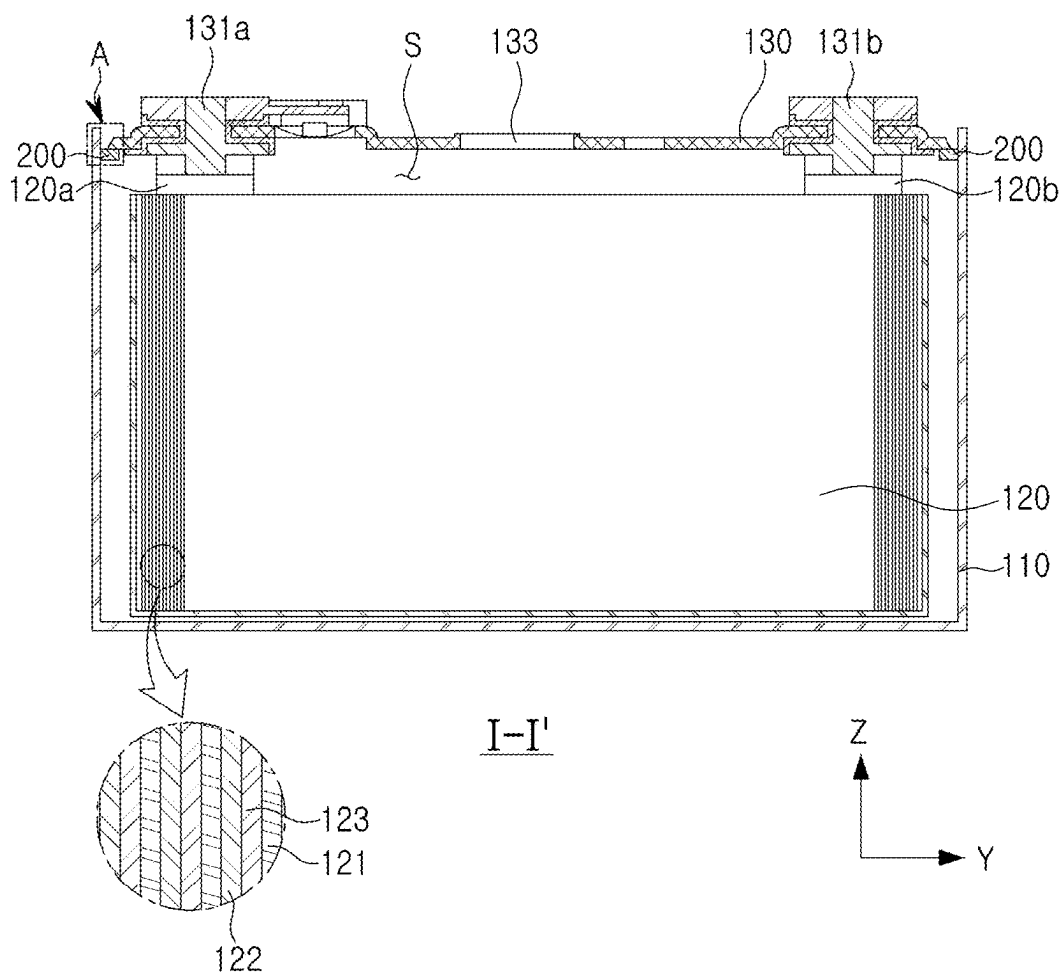


FIG. 3

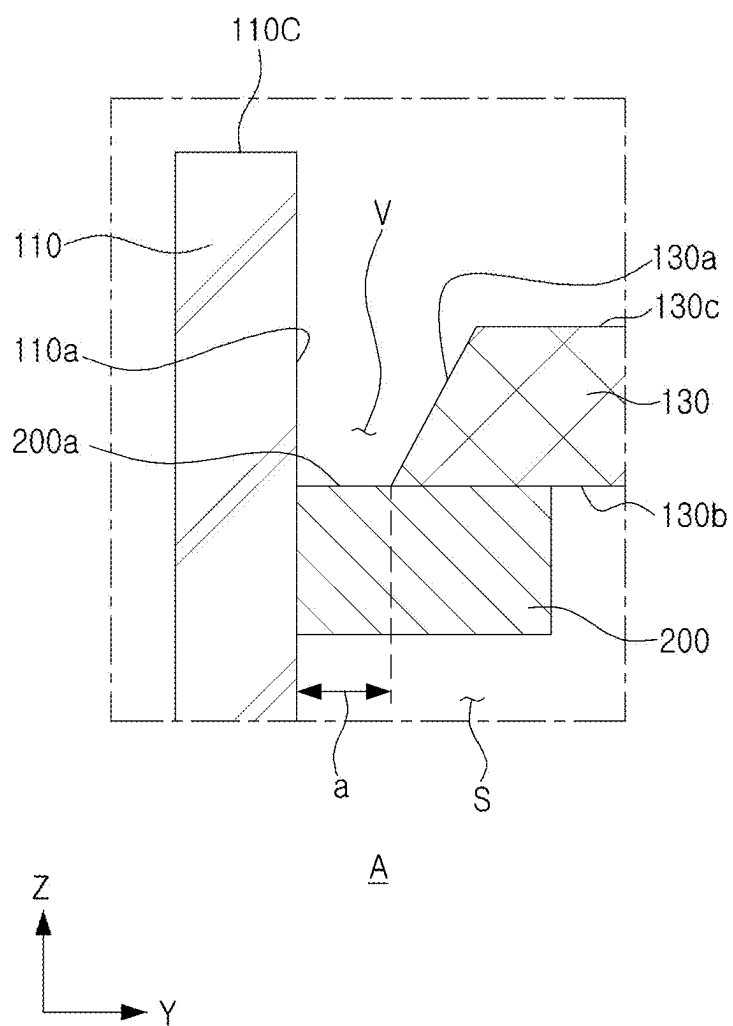


FIG. 4

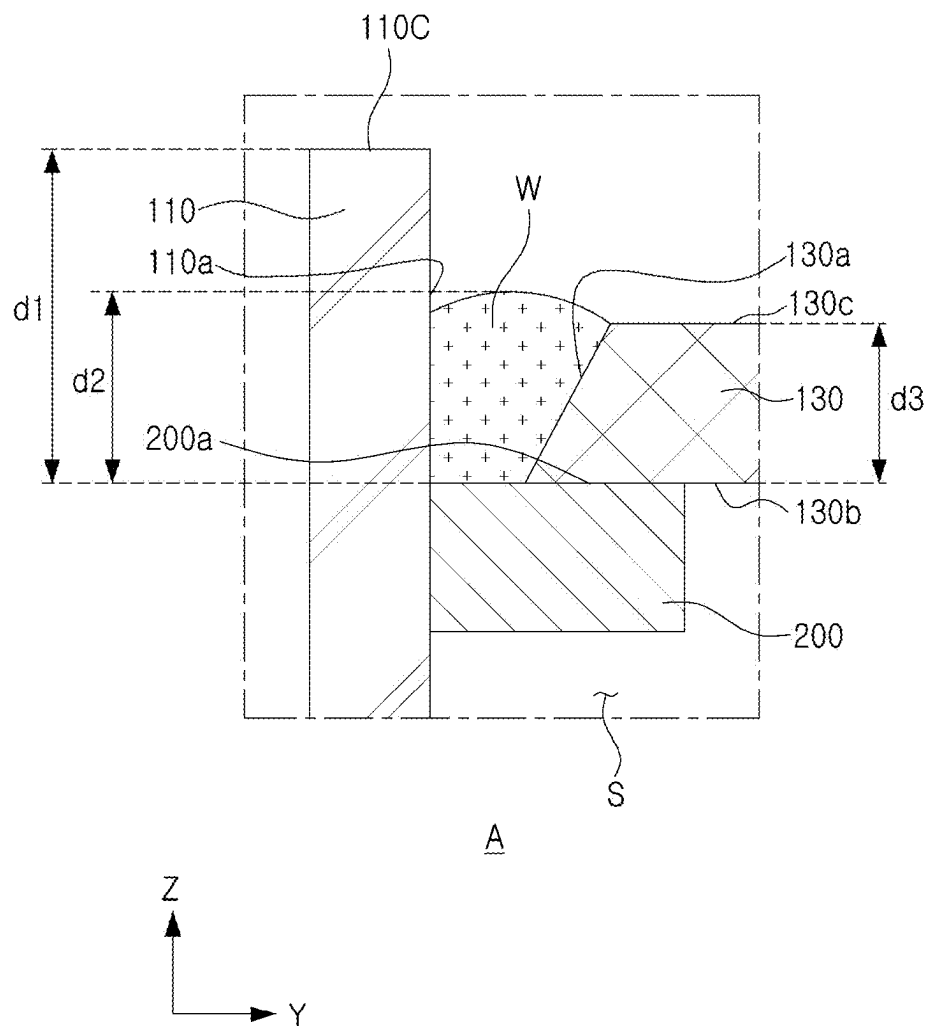


FIG. 5

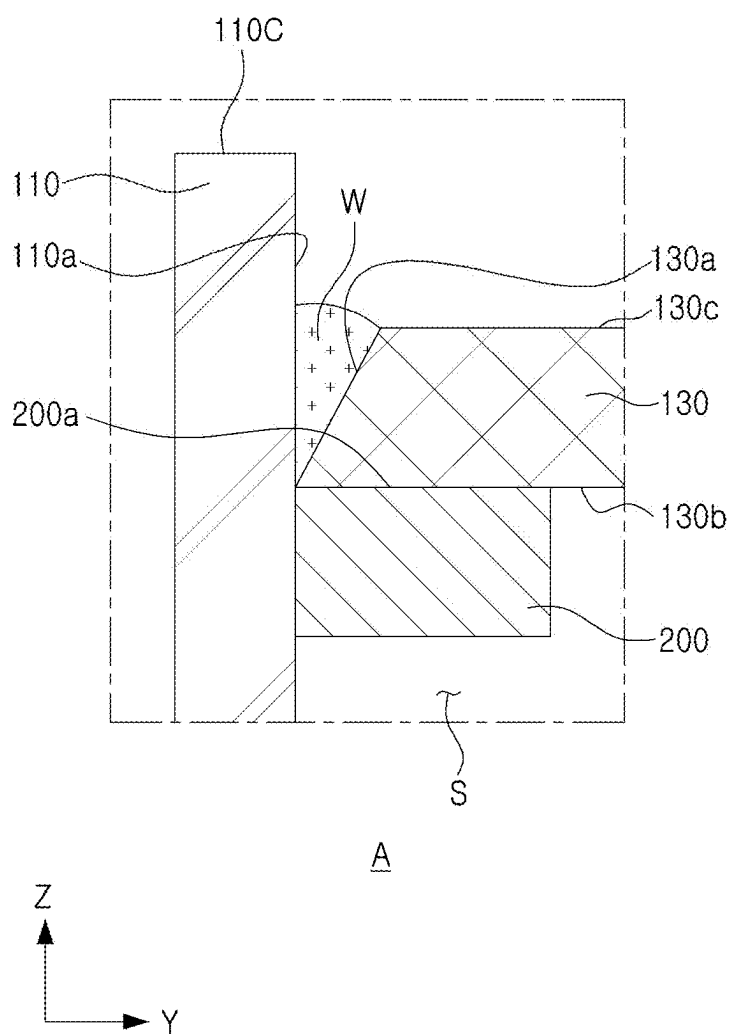


FIG. 6

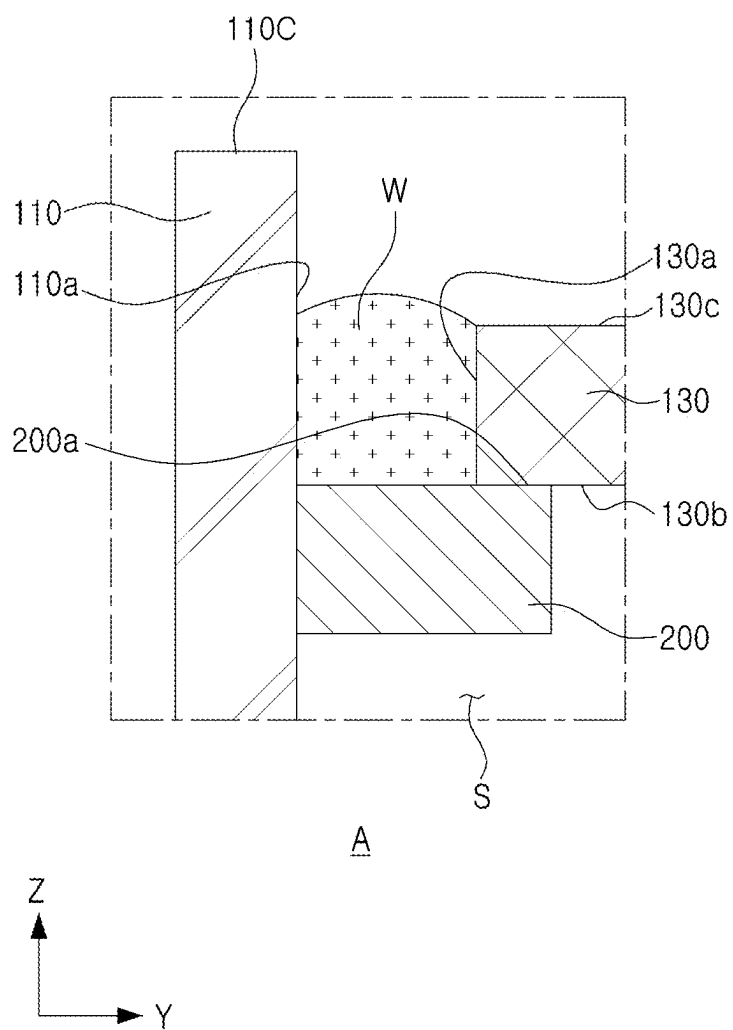


FIG. 7

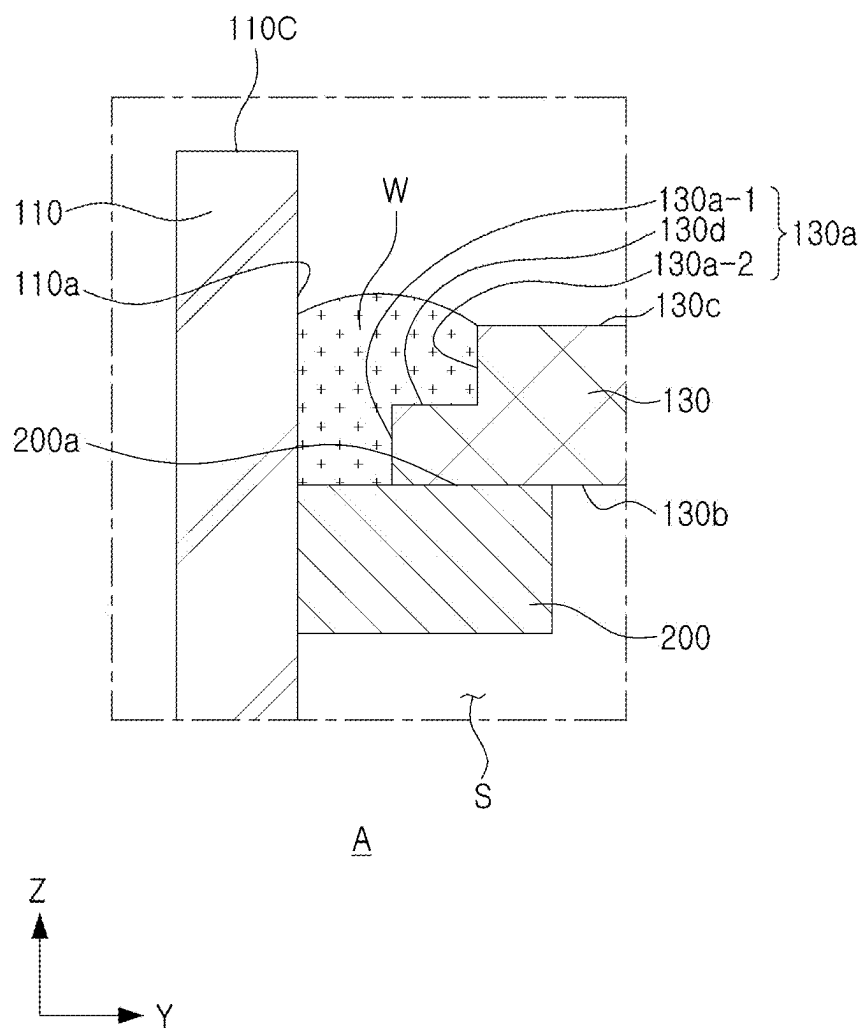


FIG. 8A

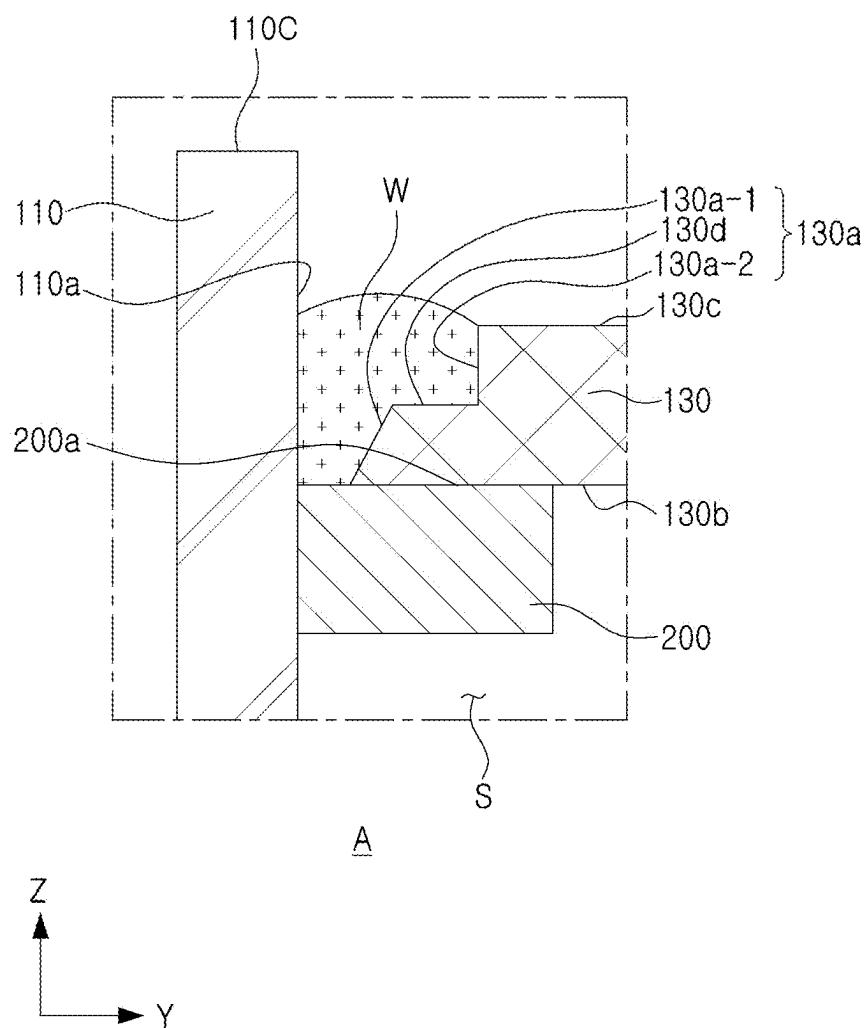


FIG. 8B

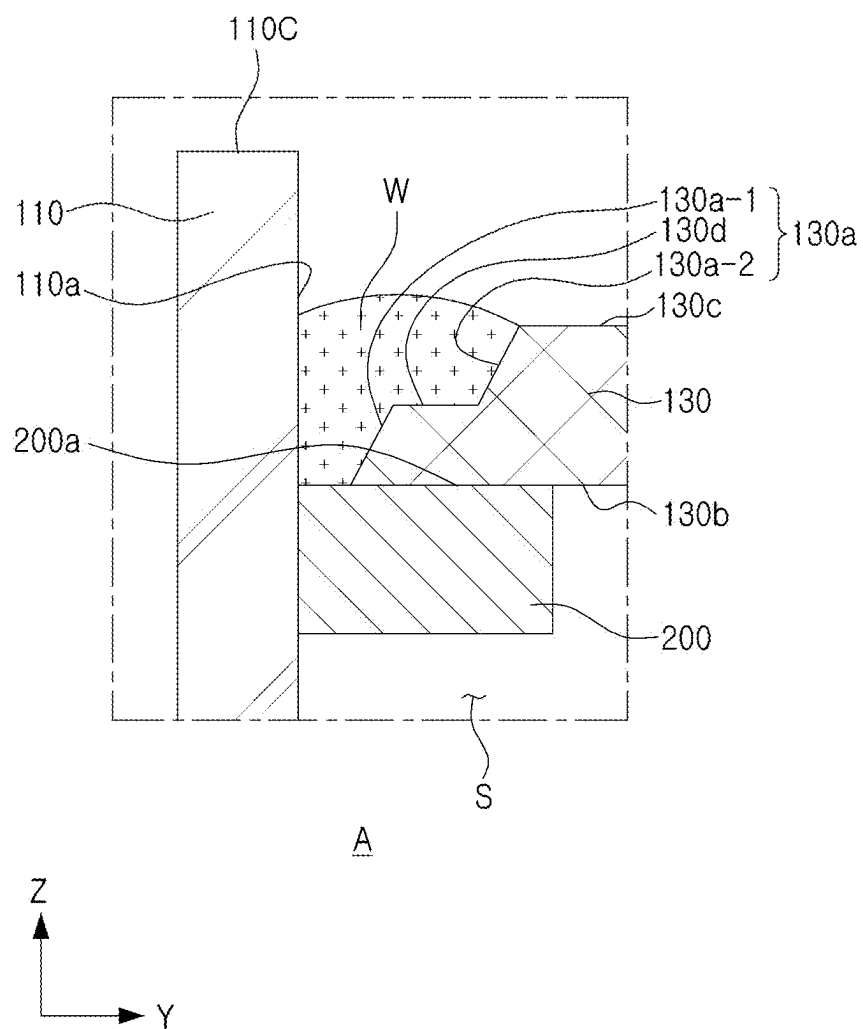


FIG. 8C

BATTERY CELL INCLUDING SUPPORT PORTION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent document claims the priority and benefits of Korean Patent Application No. 10-2024-0030521 filed on Mar. 4, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The disclosure and implementations disclosed in this patent document generally relate to a battery cell including a support portion.

BACKGROUND

[0003] Secondary batteries, unlike primary batteries, have come to prominence as a power source for various mobile devices and electric vehicles due to convenience, in that they are chargeable and dischargeable.

[0004] Secondary batteries may include battery cells in which an electrode assembly formed by stacking a positive electrode plate, a negative electrode plate, and a separator or winding them in a roll shape is accommodated inside a case together with an electrolyte. A plurality of battery cells may be stacked in a certain direction and accommodated in a battery module or battery pack.

[0005] In some types of secondary batteries, the electrode assembly may be injected into the case of the battery cell together with an electrolyte. In a state in which the electrode assembly and the electrolyte are accommodated, a cap may be coupled to an opening in the case to seal internal space of the case.

[0006] Here, the height of the battery cell may be increased due to the thickness of the cap itself or the thickness of a welding bead itself fixing the cap to the case.

[0007] The description is provided to help understand the technical background of the present disclosure and should not be construed to reduce, limit, or restrict the technical idea of the present disclosure. In addition, the contents described or suggested in the description do not necessarily mean prior art, and some may include contents that do not correspond to prior art.

SUMMARY

[0008] The present disclosure may be implemented in some embodiments to provide a battery cell having a volume reduced by a thickness of a cap plate.

[0009] According to one aspect of the present disclosure, a battery cell in which a welding bead does not protrude outwardly of a case may be provided.

[0010] In some embodiments of the present disclosure, a battery cell includes: a case accommodating an electrode assembly in an internal accommodation space and having an opening formed therein; a cap plate at least partially inserted into the accommodation space through the opening and coupled to the case; and a support portion disposed in the accommodation space and supporting a lower surface of the cap plate, wherein a side surface of the cap plate facing the internal surface of the case is spaced apart from the internal surface of the case by a predetermined distance to form a

fixing space, and the support portion is disposed below the cap plate so that at least a portion thereof forms a lower surface of the fixing space.

[0011] The battery cell may further include a welded portion at least partially disposed in the fixing space to fix the case to the cap plate, wherein the support portion supports the welded portion and the cap plate together.

[0012] The support portion may be coupled to the internal surface of the case, and at least the support portion may be provided on the internal surface of the case to support an edge of the cap plate.

[0013] The internal surface of the case may be provided as four or more surfaces, and the support portion may be provided on two or four surfaces facing each other among the internal surfaces of the case to support the cap plate.

[0014] The support portion may extend along the internal surface of the case and surround the opening of the case.

[0015] A chamfer inclined with a length increasing toward the accommodation space may be formed on the side surface of the cap plate.

[0016] The side surface of the cap plate may be provided with a step portion.

[0017] The side surface of the cap plate may include a first side surface spaced apart from the internal surface of the case; a second side surface disposed above the first side surface and having a smaller length than the first side surface; and a step portion surface connecting the first side surface to the second side surface, wherein at least one of the first side surface or the second side surface of the cap plate is formed with a chamfer inclined with a length increasing toward the accommodation space.

[0018] An upper portion of the welded portion may be disposed on the same level as or lower than an upper surface of the case.

[0019] An upper surface of the case may be disposed on the same level as or higher than an upper surface of the cap plate, and an upper portion of the welded portion may be disposed on the same level as or higher than the upper surface of the cap plate.

[0020] The upper portion of the welded portion may be disposed between the upper surface of the case and the upper surface of the cap plate.

[0021] The fixing space may be formed by the internal surface of the case, the side surface of the cap plate, and an upper surface of the support portion.

[0022] A distance between an upper surface of the support portion and an upper surface of the case may be equal to or greater than a height of the welded portion.

[0023] A length of the cap plate may be formed to be smaller than a length of the internal surface of the case.

[0024] In some embodiments of the present disclosure, a battery cell includes: a case accommodating an electrode assembly in an internal accommodation space; a cap plate at least partially inserted into the accommodation space and coupled to the case; a support portion disposed in the accommodation space to support a lower surface of the cap plate; and a welded portion disposed between and fixing an internal surface of the case and a side surface of the cap plate facing the internal surface of the case to each other, wherein the support portion is provided so that at least a portion thereof is in contact with a lower surface of the welded portion.

BRIEF DESCRIPTION OF DRAWINGS

[0025] Certain aspects, features, and advantages of the present disclosure are illustrated by the following detailed description with reference to the accompanying drawings.

[0026] FIG. 1 is a perspective view of a battery cell according to an embodiment of the present disclosure;

[0027] FIGS. 2A to 2C are exploded views of a case and a cap plate of a battery cell;

[0028] FIG. 3 is a schematic internal cross-sectional view along line I-I' of FIG. 1;

[0029] FIG. 4 is an enlarged view of A of FIG. 3;

[0030] FIG. 5 is a drawing illustrating an embodiment of the present disclosure;

[0031] FIG. 6 is a drawing illustrating another embodiment of the present disclosure;

[0032] FIG. 7 is a drawing illustrating another embodiment of the present disclosure; and

[0033] FIGS. 8A to 8C illustrate an embodiment in which a step portion structure is applied to a side surface of a cap plate.

DETAILED DESCRIPTION

[0034] Prior to the description of the present disclosure, terms and words used in the present specification and claims to be described below should not be construed as limited to ordinary or dictionary terms, and should be construed in accordance with the technical idea of the present disclosure based on the principle that the inventors may properly define their own inventions in terms of terms in order to best explain the invention.

[0035] Also, the same reference numerals or symbols respectively illustrated in the attached drawings denote parts or elements that perform the actually same functions. For convenience of description and understanding, the parts or elements will be described by using the same reference numerals or symbols even in different embodiments.

[0036] As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” etc. when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0037] In addition, in the present specification, the expressions, such as an upper side, a lower side, a side face, a rear surface, and the like, are described based on the drawings and may be expressed differently when the direction of the corresponding object is changed.

[0038] The terms including ordinal numbers, such as ‘first,’ ‘second,’ etc. may be used herein to distinguish elements from one another. These ordinal numbers are merely used to distinguish the same or similar elements from one another, and meanings of the terms are not construed as being limited by the using of the ordinal numbers. For example, use orders or arrangement orders of elements combined with these ordinal numbers are not limited by numbers thereof. The ordinal numbers may be redispersed with one another.

[0039] Hereinafter, the present disclosure will be described with reference to the drawings.

[0040] FIG. 1 is a perspective view of a battery cell according to an embodiment of the present disclosure, FIG. 2A to 2C is an exploded view of a case and a cap plate of the battery cell, and FIG. 3 is a schematic internal cross-sectional view taken along line I-I' of FIG. 1.

[0041] Referring to FIGS. 1 to 3 together, a battery cell 10 according to an embodiment of the present disclosure includes a case 110 accommodating an electrode assembly 120 in an internal accommodation space S and having an opening O, a cap plate 130 at least partially inserted into the accommodation space S through the opening O and coupled to the case 110, and a support portion 200 disposed in the accommodation space S and supporting a lower surface 130b of the cap plate 130, wherein a side surface 130a of the cap plate 130 facing an internal surface 110a of the case 110 is spaced apart from the internal surface 110a of the case 110 by a predetermined distance a to form a fixing space V, and the support portion 200 is disposed below the cap plate 130 so that at least a portion thereof forms a lower surface of the fixing space V.

[0042] In other words, the battery cell 10 of the present disclosure may include the case 110 having the opening O formed on at least one side to accommodate the electrode assembly 120 in the accommodation space S therein and the cap plate 130 accommodated at least partially in the accommodation space of the case 110 through the opening O of the case 110 and coupled to the case 110 to close the opening O. In addition, the present disclosure may include the support portion 200 disposed in the accommodation space S of the case 110 and supporting the cap plate 130 in the accommodation space S.

[0043] Meanwhile, the fixing space V may refer to a space formed by the internal surface 110a of the case 110, the side surface 130a of the cap plate 130, and an upper surface 200a of the support portion 200.

[0044] The case 110 may provide the internal space in which the electrode assembly 120 described below may be accommodated. In the drawing, the case 110 is expressed as having a roughly rectangular solid shape, but this is only an example may be changed to have various shapes as long as the accommodation space S accommodating the electrode assembly 120 is formed.

[0045] In addition, the cap plate 130 to be described below may be coupled to the opening O of the case 110 to seal the accommodation space S. A detailed structure in which the cap plate 130 is coupled to the opening O while being supported by the support portion 200 will be described below.

[0046] A material of the case 110 may be appropriately selected in consideration of thermal and electrical conductivity, rigidity corresponding to swelling of the electrode assembly 120, processability, manufacturing costs, etc. For example, the case 110 may be formed of a metallic material including aluminum, aluminum alloy, etc.

[0047] The electrode assembly 120 may be disposed in the accommodation space S of the case 110. The electrode assembly 120 may include a positive

[0048] electrode material (i.e., a cathode material) 121. The positive electrode material 121 may include a positive electrode current collector and a positive electrode active material. In some embodiments, the positive electrode current collector may include aluminum, an aluminum alloy, etc., and the positive electrode active material may include lithium cobalt oxide, lithium manganese oxide, lithium

nickel oxide, lithium iron phosphate, etc. The positive electrode active material may be coated on a surface of the positive electrode current collector. A partial region of the positive electrode current collector in which the positive electrode active material is not coated may function as a cathode tab **120a**. In some embodiments, a plurality of positive electrode tabs **120a** may be provided, and some or all of the plurality of positive electrode tabs **120a** may be mutually bonded.

[0049] The electrode assembly **120** may include a negative electrode material (i.e., an anode material) **122**. The negative electrode material **122** may include a negative electrode current collector and a negative electrode active material. In some embodiments, the negative electrode current collector may include copper, a copper alloy, nickel, a nickel alloy, etc., and the negative electrode active material may include carbon, silicon, etc. The negative electrode active material may be coated on a surface of the negative electrode current collector. A partial region of the negative electrode current collector in which the negative electrode active material is not coated may function as a negative electrode tab **120b**. In some embodiments, a plurality of negative electrode tabs **120b** may be provided, and some or all of the plurality of negative electrode tabs **120b** may be mutually bonded.

[0050] The electrode assembly **120** may include a separator **123**. The separator **123** may be disposed between the positive electrode material **121** and the negative electrode material **122**. The separator **123** may function to physically prevent the positive electrode material **121** and the negative electrode material **122** from contacting each other, while providing a passage for the migration of ions. For example, the separator **123** may include a porous polymer film or a porous non-woven fabric. In some embodiments, the separator **123** may be formed of a polymer material including polyethylene, polypropylene, etc. In addition, the separator **123** may include a dry and wet separator. In some embodiments, the separator **123** may include a coating layer including a ceramic coating layer, etc.

[0051] In the electrode assembly **120**, the above components may be arranged in a manner, such as winding or stacking. For example, the electrode assembly **120** may be formed as a structure in which the positive electrode material **121**, the negative electrode material **122**, and the separator **123** are wound around a longitudinal or transverse axis. Alternatively, the electrode assembly **120** may be formed as a structure in which the aforementioned winding structure is compressed in a direction, approximately perpendicular to the winding axis. The aforementioned winding structure may be referred to as a 'jelly roll' in the art.

[0052] As another example, the electrode assembly **120** may be formed as a structure in which the positive electrode material **121**, the negative electrode material **122**, and the separator **123** are sequentially stacked vertically (e.g., in a thickness direction or height direction of the electrode assembly). In some cases, the separator **123** in the stack structure may be formed as a structure in which a plurality of unit separators **123** continuous in a length direction are sequentially folded according to stacking of the positive electrode material **121** and the negative electrode material **122**. The aforementioned stacked structure may be referred to as 'stack and folding,' 'Z-folding,' etc. in the art. However, in the present embodiment, the arrangement of each component of the electrode assembly **120** is not particularly

limited. The electrode assembly **120** may have various arrangements other than those described above.

[0053] In some embodiments, the electrode assembly **120** may be formed by combining a plurality of block units. For example, the electrode assembly **120** may include a block unit wound in a jelly roll manner, and two or more of the block units may be combined to form the electrode assembly **120**. In the present embodiment, the electrode assembly **120** is formed by combining two jelly roll units. As another example, the electrode assembly **120** may include a block unit wound in a stack-and-fold manner, and two or more of the block units may be combined to form the electrode assembly **120**.

[0054] The electrode assembly **120** may be accommodated in the accommodation space **S** of the case **110** together with the electrolyte. In some embodiments, the electrolyte may be formed of an organic solvent including a lithium salt. For example, the lithium salt may include lithium hexafluorophosphate (LiPF₆), lithium tetrafluoroborate (LiBF₄), etc. in a liquid or gel state, and the organic solvent may include a cyclic carbonate, such as ethylene carbonate (EC), propylene carbonate (PC), etc. or a linear carbonate, such as diethyl carbonate (DEC), dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), etc.

[0055] In some other embodiments, the electrolyte may be omitted or substituted. For example, when an inorganic solid electrolyte is used, the liquid or gel electrolyte may be omitted.

[0056] In addition, the structure of the electrode assembly **120** illustrated in the present disclosure is an example. In the present disclosure, the electrode assembly **120** including the positive electrode tab **120a** and the negative electrode tab **120b** arranged in parallel from one side of the electrode assembly **120** is illustrated, but the positions and/or numbers of the positive electrode tab **120a** and the negative electrode tab **120b** are examples. For example, in an embodiment not illustrated, the positive electrode tab **120a** and the negative electrode tab **120b** may be located with the electrode assembly **120** (e.g., the positive electrode material **121**, the negative electrode material **122**, and the separator **123**) interposed therebetween. At least a portion of the positive electrode tab **120a** and the negative electrode tab **120b** may be located between the case **110** and the electrode assembly **120**, respectively.

[0057] The cap plate **130** may be coupled to the case **110** to close the opening of the case **110**. In the drawing, the cap plate **130** is illustrated as a quadrangular plate shape corresponding to the opening of the case **110**. The cap plate **130** may be coupled to the case **110** to close the accommodation space **S** of the case **110** in which the electrode assembly **120** is disposed. In some embodiments, the cap plate **130** may be welded to the case **110** by ultrasonic welding, laser welding, or the like.

[0058] A length of the cap plate **130** may be formed to be smaller than a length of the internal surface **110a** of the case **110**. Through this, the cap plate **130** may be easily inserted into the opening **O**. Meanwhile, in the present disclosure, the term "length" encompasses at least one of length, width or diameter, and is not necessarily limited to the length of a circle.

[0059] Terminal portions **131a** and **131b** electrically connected to the electrode assembly **120** may be arranged on the cap plate **130**. The battery cell **10** may be electrically connected to an external power source through the terminal

portions **131a** and **131b**. The terminal portions **131a** and **131b** may include a positive electrode terminal **131a** and a negative electrode terminal **131b**. The positive electrode terminal **131a** may be electrically connected to the positive electrode tab **120a** of the electrode assembly **120**, and the negative electrode terminal **131b** may be electrically connected to the negative electrode tab **120b** of the electrode assembly **120**.

[0060] The cap plate **130** may include a vent **133**. In the present embodiment, the vent **133** is disposed between the positive electrode terminal **131a** and the negative electrode terminal **131b**. However, the position of the vent **133** may vary as needed and is not necessarily limited to the example. In some other embodiments, a vent **133** may be disposed on or added to the case **110**. The vent **133** may be formed to be opened in response to internal pressure of the case **110**. The vent **133** may function to discharge the internal pressure to the outside of the case **110** to contribute to stabilizing the internal elements of the case **110**. In the present embodiment, the vent **133** may include a notch (not shown) having a predetermined shape. The notch may be a vulnerable portion formed to be at least partially sunken in the thickness direction of the vent **133**. By inducing the vent **133** to be fractured according to the internal pressure *e* of the accommodation space *S* through the notch, the vent **133** may be opened to discharge a fluid (gas, etc.) in the accommodation space *S* of the case **110** to the outside.

[0061] Meanwhile, in the present disclosure, the secondary battery including the case **110** including one opening **0** and the cap plate **130** located on one side of the case **110** is illustrated, but this is an example. In an embodiment not illustrated, the secondary battery including openings **O** formed on both sides of the case **110** and a plurality of cap plates **130** respectively located in the openings **O** may be provided. For example, the secondary battery may include a first cap plate located on one side of the case **110** and a second cap plate located on the other side of the case **110**.

[0062] The support portion **200** of the present disclosure may be located to be close to the opening **O** in the accommodation space *S*. The support portion **200** may support a lower surface of the edge of the cap plate **130** at least partially inserted into the accommodation space *S*. That is, the support portion **200** is coupled to the internal surface **110a** of the case **110**, and at least one support portion **200** may be provided on the internal surface **110a** of the case **110** to support the edge of the cap plate **130**.

[0063] For example, as shown in FIGS. 2A and 2B, the support portions **200** may be arranged at two or four corners facing each other in the accommodation space *S* to support the corner portions of the edges of the cap plate **130**. That is, the support portions **200** may be arranged at two or four corners facing each other on the internal surface of the case **110** to support the cap plate **200**.

[0064] In addition, the support portions **200** may be provided to support the four corners of the cap plate **130**. Alternatively, the support portions **200** may be arranged at two of the four corners to support the cap plate **130**. In other words, the support portion **200** may be arranged at two or four edge surfaces of the internal surface **110a** of the case **110** facing each other to support the cap plate **130**.

[0065] In other words, referring to FIGS. 2A and 2B, in an embodiment of the present disclosure, the internal surface **110a** of the case **110** is provided in four or more (illustrated as four in the drawing) and the support portions **200** may be

provided in an even number and may be arranged on an even number of the internal surfaces of the case **110**. Here, in order to support the cap plate **130**, the support portions **200** may be arranged on the internal surfaces **110a** facing each other.

[0066] That is, according to an embodiment of the present disclosure, four or more internal surfaces **110a** of the case **110** may be provided, and the support portion **200** may be provided on two or four of the internal surfaces **110a** of the case **110** to face each other. FIG. 2A illustrates that the support portion **200** is provided on four surfaces, and FIG. 2B illustrates that the support portion **200** is provided on two surfaces. In addition, as illustrated in FIG. 2A to 2B, the support portion **200** may be formed to surround the internal surface (**110a**, see FIG. 4) of the case **110** to fully support the edge of the cap plate **130**. In other words, the support portion **200** may be formed to extend along the internal surface **110a** of the case **110** to surround the opening **O** of the case **110**. Through this, the support portion **200** may support the entire edge of the cap plate **130**.

[0067] In this manner, the support portion **200** of the present disclosure is not limited thereto and may have any structure capable of supporting a welded portion *W* (see FIG. 5) to be described below and the cap plate **130** together.

[0068] Hereinafter, the coupling structure of the case **110** and the cap plate **130** through the support portion **200** of the present disclosure will be described in detail.

[0069] FIG. 4 is an enlarged view of A of FIG. 2A to 2C, and FIG. 5 is a drawing illustrating an embodiment of the present disclosure. FIG. 4 and FIG. 5 are enlarged views of the coupling structure of the cap plate **130** and the case **110**.

[0070] Referring to FIGS. 4 and 5 together, the battery cell **10** according to an embodiment of the present disclosure may include the support portion **200** disposed in the accommodation space *S* to support the cap plate **130** at least partially inserted into the accommodation space *S*.

[0071] More specifically, according to an embodiment, the cap plate **130** may be at least partially accommodated in the accommodation space *S* through the opening **O** of the case **110**. Here, “at least partially accommodated” may mean that at least a portion of the lower surface **130b** of the cap plate **130** (for example, an edge portion of the cap plate **130**) is disposed to be lower (in a *-Z*-axis direction) than the upper surface **110c** of the case **110**.

[0072] In an embodiment, the fixing space *V* may be formed between the internal surface **110a** of the case **110** and the side surface **130a** of the cap plate **130**. The fixing space *V*, a space between the internal surface **110a** of the case **110** and the side surface **130a** of the cap plate **130**, may be a portion in which the welded portion *W* to be described below is formed.

[0073] For example, the side surface **130a** of the cap plate **130** facing the internal surface **110a** forming the accommodation space *S* of the case **110** may be spaced apart from the internal surface **110a** of the case **110** by the predetermined distance *a* to form the fixing space *V*. In addition, by separating the cap plate **130** and the case **110** in this manner, it is possible to prevent or minimize stress from being concentrated on the edge (the side surface; portion in which **130a** is provided) of the cap plate **130**. For example, a chamfer may be formed on the side surface **130a** of the cap plate **130** so that the fixing space *V* widens upwards (in a *+Z*-axis direction). In other words, the cap plate **130** may have a chamfer formed to be inclined on the side surface

130a so that the length thereof increases toward the accommodation space **S**. That is, the side surface **130a** of the cap plate **130** may be provided to be inclined to have the length increasing toward the accommodation space **S**.

[0074] The drawing illustrates a structure in which a chamfer is formed on the side surface **130a** of the cap plate **130**.

[0075] However, this is only an embodiment, and any structure allowing the fixing space **V** to be formed may be employed even without a chamfer. More specifically, the present disclosure may include at least one of a first structure (see FIG. 7) in which the side surface **130a** of the cap plate **130** is spaced apart from the case **110** by the predetermined distance **a** so that the fixing space **V** is formed, a second structure in which a chamfer is formed on the side surface **130a** of the cap plate **130**, a third structure (see FIG. 8A to 8C) in which the side surface **130a** of the cap plate **130** is formed stepwise, or a fourth structure in which both the first structure and the second structure are applied. FIGS. 4 and 5 are described based on the fourth structure in which both the first and second structures described above are applied. As will be described below, FIG. 6 is based on an embodiment in which only the second structure is applied, FIG. 7 is based on an embodiment in which the first structure is applied, and FIG. 8A to 8C is based on an embodiment in which the third structure is applied.

[0076] As such, the present disclosure is not particularly limited as long as the fixing space **V** is formed between the case **110** and the cap plate **130** and the support portion **200** described below supports the lower surface of the fixing space **V**.

[0077] The support portion **200** may be coupled to the internal surface **110a** of the case **110** and may be arranged in the accommodation space **S**. The upper surface **200a** of the support portion **200** may be disposed to be closer to the lower surface (the $-Z$ -axis direction) than the upper surface **110c** of the case **110**. The support portion **200** may be disposed to be lower (the $-Z$ -axis direction) than the cap plate **130** in the accommodation space **S** to support the cap plate **130**.

[0078] In addition, the support portion **200** may form a lower surface of the fixing space **V** formed between the internal surface **110a** of the case **110** and the side surface **130a** of the cap plate **130**.

[0079] Here, “forming the lower surface of the fixing space **V**” may be disposed to support a lower surface of the welded portion **W** formed in the fixing space **V** to be described below. That is, the support portion **200** may be provided to support not only the lower surface **130b** of the cap plate **130** but also the lower surface of the welded portion **W** to be described below.

[0080] Referring to FIG. 5, the welded portion **W** may be formed in the fixing space **V**. The welded portion **W** may fix the internal surface **110a** of the case **110** to the side surface of the cap plate **130a**.

[0081] In other words, the battery cell **10** according to an embodiment of the present disclosure may further include the welded portion **W** at least partly disposed in the fixing space **V** to fix the case **110** to the cap plate **130**. Here, the support portion **200** may be provided to support the welded portion **W** and the cap plate **130** together. Here, “supporting together” may mean that the upper surface **200a** of the support portion **200** contacts the lower surface of the welded portion **W** and the lower surface **130b** of the cap plate **130**

to support them together. That is, the support portion **200** forming the lower surface of the fixing space **V** may support the lower surface of the welded portion **W**. In addition, “forming the lower surface of the fixing space **V**” may mean that the support portion **200** is provided to support the lower surface of the welded portion **W**.

[0082] In other words, according to an embodiment of the present disclosure, the battery cell **10** may include the case **110** accommodating the electrode assembly **120** in the internal accommodation space **S**, the cap plate **130** at least partially inserted into the accommodation space **S** and coupled to the case **110**, the support portion **200** disposed in the accommodation space **S** to support the lower surface **130b** of the cap plate **130**, and the welded portion **W** disposed between the internal surface **110a** of the case **110** and the side surface **130a** of the cap plate **130** facing the internal surface **110a** of the case **110** to fix them to each other, wherein the support portion **200** may be provided such that at least a portion thereof is in contact with the lower surface of the welded portion **W**.

[0083] Meanwhile, in an embodiment, the welded portion **W** may be a welding bead provided to fix the internal surface **110a** of the case **110** and the side surface **130a** of the cap plate **130** to each other. For example, the welded portion **W** may be formed by melting at least one of the case **110** or the cap plate **130** by irradiating the vicinity of the fixing space **V** with a laser by a method, such as laser welding. The lower surface of the welded portion **W** may be supported by the support portion **200**. That is, the welded portion **W** may fix the case **110**, the cap plate **130**, and the support portion **200** together. An upper portion (the most protruding portion in the $+Z$ -axis direction) of the welded portion **W** may be located at the same height as the upper surface **110c** of the case **110** or lower (in the $-Z$ -axis direction). In other words, referring to FIG. 5, a height **d2** of the welded portion **W** may be equal to or less than a distance **d1** between the support **200** and the case **110**.

[0084] That is, the distance **d1** between the upper surface **200a** of the support **200** and the upper surface **110a** of the case **110** may be equal to or greater than the height **d2** of the welded portion **W**.

[0085] Here, the height of the welded portion **W** may refer to a vertical distance from the lower surface of the welded portion **W** to the top of the welded portion **W** or a vertical distance from the upper surface **200a** of the support **200** to the top of the welded portion **W**.

[0086] More specifically, in an embodiment, the size of the welded portion **W** may increase depending on the temperature of heat based on laser irradiation. If at least a portion of the welded portion **W** is located above the upper surface **110c** of the case **110** (if the welded portion **W** is located outside the case), the size of the battery cell **10** may increase. In addition, external impact may be concentrated on the protruding welded portion **W**, making it more vulnerable to the impact and reducing the durability of the battery cell **10**. Accordingly, according to an embodiment of the present disclosure, the welded portion **W** may be disposed in the accommodation space **S** of the case **110**.

[0087] That is, the upper surface **110c** of the case **110** may be disposed on the same level as or higher than the upper surface **130c** of the cap plate **130**, and the upper portion of the welded portion **W** may be disposed on the same level as or lower (in the $-Z$ -axis direction) than the upper surface **110c** of the case **110**. Specifically, the distance **d1** between

the upper surface **200a** of the support **200** and the upper surface **110a** of the case may be equal to or greater than a distance **d3** between the upper surface **200a** of the support portion and the upper surface **130c** of the cap plate. In addition, the distance **d3** between the upper surface **200a** of the support portion and the upper surface **130c** of the cap plate may be equal to or smaller than the height **d2** of the welded portion **W**.

[0088] The welded portion **W** may be disposed to completely fill the fixing space **V** for bonding force between the case **110** and the cap plate **130**. In an embodiment, the upper portion of the welded portion **W** may be disposed as the same level as or higher (in the +Z-axis direction) than the upper surface **130c** of the cap plate **130**.

[0089] For example, the upper portion of the welded portion **W** may be disposed between the upper surface **110c** of the case **110** and the cap plate **130c**.

[0090] In this manner, in an embodiment, the upper portion of the welded portion **W** may be disposed on the same level as or lower (in the -Z-axis direction) than the upper surface **110c** of the case **110**, but may be disposed on the same level as or higher than the upper surface **130c** of the cap plate **130**.

[0091] Meanwhile, in an embodiment, the upper surface **130c** of the cap plate **130** may be disposed on the same level as or lower (in the -Z-axis direction) than the upper surface **110c** of the case **110**. Through the aforementioned structure, the height of the battery cell **10** may be reduced according to the present disclosure.

[0092] Meanwhile, referring to FIG. 2A to 2C together, in an embodiment, the terminal portions **131a** and **131b** electrically connected to an external power source may be disposed on the cap plate **130**, but an upper surface thereof may be disposed on the same level as or higher than the upper surface **110c** of the case **110**. In other words, in an embodiment, the upper surface **110c** of the case **110** may be disposed between the upper surface of the terminal portions **131a** and **131b** and the upper surface of the cap plate **130c** based on the height direction (the Z-axis direction).

[0093] However, the present disclosure is not limited thereto, and any structure in which at least a portion of the cap plate **130** is inserted into the accommodation space **S** so that the height of the battery cell **10** is reduced may be included in the present disclosure.

[0094] FIG. 6 is a drawing illustrating another embodiment of the present disclosure, in which the second structure (only a chamfer is formed on the side surface **130a** of the cap plate) is illustrated.

[0095] According to another embodiment, a chamfer may be formed on the side surface **130c** of the cap plate **130** to form a predetermined space (the 'fixing space; V') together with the internal surface **110a** of the case **110**, and the welded portion **W** may be formed in the fixing space **V**.

[0096] In the present embodiment, a lower surface of the welded portion **W** may be supported by the side surface **130c** of the cap plate **130**, and the support portion **200** may be disposed to face the welded portion **W** in the height direction (the Z-axis direction) with the cap plate **130** interposed therebetween.

[0097] Meanwhile, another embodiment of the present disclosure will be described with reference to FIGS. 7 and 8.

[0098] FIG. 7 illustrates another embodiment of the present disclosure. Referring to FIG. 7, the side surface **130a** of

the cap plate **130** according to an embodiment of the present disclosure may be provided to face the internal surface **110a** of the case **110** in parallel.

[0099] FIGS. 8A to 8C illustrate an embodiment in which a step portion structure is applied to the side surface of the cap plate. Referring to FIGS. 8A to 8C, the side surface **130a** of the cap plate **130** may be provided with a step portion. Through the step portion structure, a contact area between the welded portion **W** and the cap plate **130** may be further increased.

[0100] In addition, the chamfer structure described above may of course be applied to the step portion structure as well.

[0101] More specifically, the side surface **130a** of the cap plate **130** may include a first side surface **130a-1** spaced apart from the internal surface **110a** of the case **110**, a second side surface **130a-2** disposed above the first side surface **130a-1** and having a smaller length than the first side surface, and a step portion surface **130d** connecting the first side surface **130a-1** to the second side surface **130a-2**. Here, a chamfer may be formed on at least one of the first side surface **130a-1** or the second side surface **130a-2** of the cap plate, the chamfer being formed in an inclined shape with a length increasing toward the accommodation space **S**.

[0102] For example, FIG. 8B illustrates a structure in which a chamfer is formed on the first side surface **130a-1**, and FIG. 8C illustrates a structure in which chamfers are formed on both the first side surface **130a-1** and the second side surface **130a-2**, respectively.

[0103] In addition, although FIGS. 8A to 8C illustrate that the side surface **130a** is formed as a two-stage step portion structure, the present disclosure is not limited thereto. In other words, any step portion structure in which the side surface **130a** is applied, even if it being formed as a three-stage or more step portion structure, will all fall within the present disclosure. Although various embodiments of the present disclosure have been described in detail above, the scope of the present disclosure is not limited thereto and it will be obvious to those skilled in the art that various modifications and variations may be made without departing from the scope of the present disclosure described in the claims. Some components of the aforementioned embodiments may be deleted and each embodiment may be implemented in combination with each other.

[0104] In the battery cell according to an embodiment of the present disclosure, the volume thereof may be reduced by the thickness of the cap plate.

[0105] In the battery cell according to an embodiment of the present disclosure, the welding bead may not protrude outside the case.

[0106] The description is merely an example of applying the principles of the present disclosure, and other components may be included without departing from the scope of the present disclosure.

[0107] Only specific examples of implementations of certain embodiments are described. Variations, improvements and enhancements of the disclosed embodiments and other embodiments may be made based on the disclosure of this patent document.

What is claimed is:

1. A battery cell comprising:

a case accommodating an electrode assembly in an internal accommodation space and having an opening formed therein;

a cap plate at least partially inserted into the accommodation space through the opening; and

a support portion disposed in the accommodation space and supporting a lower surface of the cap plate,

wherein a side surface of the cap plate facing the internal surface of the case is spaced apart from the internal surface of the case by a predetermined distance to form a fixing space, and the support portion is disposed below the cap plate so that at least a portion thereof forms a lower surface of the fixing space.

2. The battery cell of claim 1, further comprising:

a welded portion at least partially disposed in the fixing space to fix the case to the cap plate,

wherein the support portion supports the welded portion and the cap plate together.

3. The battery cell of claim 1, wherein the support portion is coupled to the internal surface of the case, and at least one of the support portion is provided on the internal surface of the case to support an edge of the cap plate.

4. The battery cell of claim 3, wherein the internal surface of the case is provided as four or more surfaces, and the support portion is provided on two or four surfaces facing each other among the internal surfaces of the case to support the cap plate.

5. The battery cell of claim 1, wherein the support portion extends along the internal surface of the case and surrounds the opening of the case.

6. The battery cell of claim 1, wherein a chamfer inclined with a length increasing toward the accommodation space is formed on the side surface of the cap plate.

7. The battery cell of claim 2, wherein the side surface of the cap plate is provided with a step portion.

8. The battery cell of claim 7, wherein

the side surface of the cap plate includes:

a first side surface spaced apart from the internal surface of the case;

a second side surface disposed above the first side surface and having a smaller length than the first side surface; and

a step portion surface connecting a first side surface to the second side surface,

wherein at least one of the first side surface or the second side surface of the cap plate is formed with a chamfer inclined with a length increasing toward the accommodation space.

9. The battery cell of claim 1, wherein an upper portion of the welded portion is disposed on the same level as or lower than an upper surface of the case.

10. The battery cell of claim 2, wherein an upper surface of the case is disposed on the same level as or higher than an upper surface of the cap plate, and an upper portion of the welded portion is disposed on the same level as or higher than the upper surface of the cap plate.

11. The battery cell of claim 10, wherein the upper portion of the welded portion is disposed between the upper surface of the case and the upper surface of the cap plate.

12. The battery cell of claim 2, wherein the fixing space is formed by the internal surface of the case, the side surface of the cap plate, and an upper surface of the support portion.

13. The battery cell of claim 2, wherein a distance between an upper surface of the support portion and an upper surface of the case is equal to or greater than a height of the welded portion.

14. The battery cell of claim 1, wherein a length of the cap plate is formed to be smaller than a length of the internal surface of the case.

15. A battery cell comprising:

a case accommodating an electrode assembly in an internal accommodation space;

a cap plate at least partially inserted into the accommodation space and coupled to the case;

a support portion disposed in the accommodation space to support a lower surface of the cap plate; and

a welded portion disposed between and fixing an internal surface of the case and a side surface of the cap plate facing the internal surface of the case to each other,

wherein at least a portion of the support portion is in contact with a lower surface of the welded portion.

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